

Cboe VIX Index — methodology & term structure

Source: Cboe Global Markets — VIX hub, methodology whitepaper & term-structure indices — https://www.cboe.com/tradable_products/vix/

What it is

The Cboe Volatility Index (VIX) is the market's benchmark gauge of **30-day expected volatility of the S&P 500**, distilled from SPX index-option prices. It is quoted in annualized percentage points (e.g. VIX = 20 implies a ~20% annualized standard deviation of SPX returns, roughly $\pm 5.8\%$ over the coming month). Crucially, it is **not** the implied volatility of any single option: it is a **model-free** synthesis of the *entire* SPX option chain. Introduced in its current form in 2003 (Cboe with Goldman Sachs), it is the "premier barometer of investor sentiment," widely read as the market's fear gauge.

Core concepts

- **Model-free implied volatility.** The VIX price-averages weighted out-of-the-money SPX/SPXW puts and calls across a wide strip of strikes, adapting the model-free variance theory of Britten-Jones & Neuberger (2000) and the replication logic of Demeterfi et al. (1999). No Black-Scholes assumption is needed.
- **Constant 30-day maturity.** Two expirations bracketing 30 days (near-term <30d, next-term >30d) are each converted to a variance, then **linearly interpolated** to a rolling 30-day horizon.
- **Mean reversion.** Volatility spikes on stress and decays toward a long-run average of ~19-20. This decay drives the shape of the VIX futures term structure.
- **Negative correlation with SPX.** Mathematically VIX has no directional bias, but it usually rises as SPX falls (leverage effect + panic put demand), making it a hedge-like, contrarian signal.

Key metrics/signals it defines + how to read/compute them

- **VIX level → regime bucket.** Rule-of-thumb bands: **calm <15**, **normal ~15-20**, **elevated ~20-30**, **stress >30**, **panic/crisis >40** (2008 peaked ~80.9). Extremes (>30) historically precede the best forward equity returns; <13 flags complacency.
- **Computation inputs.** Market prices of SPX + SPXW options, plus U.S. Treasury yields as the risk-free rate; a forward index level **F**, the first OTM strike **K**, and per-strike contributions $\Delta K/K^2 \cdot e^{rT} \cdot Q(K)$.
Variance → volatility = $\sqrt{\text{variance}} \times 100$.
- **Term structure (VIX9D → VIX → VIX3M → VIX6M):** same methodology at 9, 30, 93, 184 days — a volatility "yield curve." **Contango** (upward slope, VIX9D far) = acute stress; the ****VIX/VIX3M ratio >1.0**** flags inversion, ****>1.10**** = deep backwardation/dislocation. Curve inversion is a more reliable crisis signal than VIX level alone.

How it maps to the Trader Screener

- **§2f "VIX" (Macro context, [free] FRED-collected).** Ingest FRED series **VIXCLS** (daily close, history since 1990) and derive a **regime column** by bucketing the level (calm/normal/elevated/stress/panic). Use as a Filter (e.g. suppress new longs when VIX >30) and Display-only context banner. This is the highest-value, zero-cost mapping.
- **§2h "VIX / macro vol" (Options/derivatives signals).** The same VIXCLS feed populates the options tab's regime filter — the spec explicitly notes it "shares FRED macro tab." Lets options-oriented signals be gated by market-wide vol without any paid options feed.
- **§2h "IV term structure (contango/backwardation)." Compute a **curve-sign flag** from VIX vs VIX3M (and VIX9D vs VIX) as a proxy for near- vs far-dated event risk — a red "backwardation/stress" flag column when the ratio crosses 1.0/1.10. Cheaper than a per-name option chain yet captures the same near-vs-far signal.**
- **§2f "Rates / yield curve" + §2g risk/sizing.** Blend the VIX regime with the yield-curve regime into a composite macro filter; scale **ATR-based position size** and **stop distance** down as the regime worsens.

Actionable takeaways

- Add **VIXCLS** to the nightly FRED pull and expose a **regime column + composite macro banner** — a P1, free win.
- Store **VIX9D/VIX3M** too so the screener can flag **term-structure inversion**, a stronger crisis tell than level alone.
- Treat VIX as **contrarian/regime context**, not a stock filter: tighten sizing and demand higher conviction in stress; harvest quality names when panic (>30-40) mean-reverts.

Open questions

- Does FRED carry VIX9D/VIX3M/VIX6M, or must those come from Cboe/yfinance for the term-structure flag?
- Exact regime thresholds — fixed bands vs a rolling **VIX percentile** normalizer (parallels §2h "IV Rank/percentile")?
- Should the regime gate longs outright, or only rescale §2g position size and §2d composite score?

Sources

- Cboe — VIX Index hub
- S&P DJI — A Practitioner's Guide to Reading VIX (whitepaper)
- Cboe — Volatility Index Methodology (PDF)

- [SimTrade](#) — CBOE Volatility Index (methodology & interpretation)
- [FlashAlpha](#) — VIX Term Structure
- [FRED](#) — CBOE Volatility Index: VIX (VIXCLS)